

# Standard Operating Procedure: Python and Jupyter Notebook Installation Guide

KPMG Advisory | Delivery & Technology Enablement

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## 1. Executive Summary & Objective

This Standard Operating Procedure (SOP) outlines the standardized, secure method for installing and configuring a local Python development environment tailored for Jupyter Notebook execution on KPMG-managed workstations. Standardizing our data science and analytical tooling ensures code reproducibility, compliance with firm security controls, and seamless integration with firm-approved data pipelines and client engagement workflows.

## 2. Prerequisites & Enterprise Pre-Checks

Before proceeding with installation, ensure your workstation meets the following enterprise requirements:

- Operating System: Windows 11 Enterprise (64-bit) / macOS Sonoma or later (KPMG Enterprise Build).
- Administrative Rights: Standard user privileges (or temporary CyberArk/Privilege Management elevation if local software installation policy requires).
- Corporate Network / VPN: GlobalProtect or Zscaler active connection.
- Hardware Minimums: 16 GB RAM (32 GB recommended for advanced modeling), minimum 10 GB available SSD storage.

## 3. Recommended Installation Path: Anaconda Distribution / Miniconda

For KPMG data analytics and advisory workflows, the Anaconda / Miniconda Distribution is the primary enterprise-recommended approach due to its robust environment management and pre-packaged scientific libraries.

### Method A: Miniconda Setup (Lightweight & Recommended)

1. Download Installer: Access the official KPMG Software Portal or approved repository and download the latest 64-bit Miniconda installer for Python 3.11+.
2. Installation Wizard:
  - Run the installer with default settings.
  - Select "Just Me (recommended)" for destination folder (typically C:\Users\<YourUser>\Miniconda3).
  - Check the box "Register Miniconda3 as my default Python".
3. Initialize Environment via Terminal:  
Open the Anaconda Prompt or standard PowerShell and execute:

```
conda update -n base -c defaults conda -y
```

## Method B: Standard Python (Vanilla Python 3.11/3.12)

If using standard Python from python.org:

1. Run the official installer executable.
2. Ensure "Add python.exe to PATH" is selected.
3. Select "Install Now" or customize to  
C:\Users\<YourUser>\AppData\Local\Programs\Python\Python311.

## 4. Enterprise Proxy & SSL Certificate Configuration

When installing packages behind KPMG network firewalls and proxy appliances (Zscaler), pip and conda require proxy awareness or corporate root certificates.

### Configuring pip with Enterprise Certificates

Create or edit the configuration file at %APPDATA%\pip\pip.ini (Windows) or ~/.pip/pip.conf (macOS/Linux):

```
[global]
trusted-host = pypi.org
               files.pythonhosted.org
               pypi.python.org
timeout = 60
```

## 5. Setting Up Isolated Virtual Environments

To ensure project isolation and prevent dependency conflicts across client engagements, always create a dedicated virtual environment.

### Creating a Conda Environment:

```
# Create a dedicated environment named 'kpmg_analytics'
conda create -n kpmg_analytics python=3.11 -y

# Activate the environment
conda activate kpmg_analytics
```

### Alternative: Python venv:

```
# Create virtual environment in your project directory
python -m venv .venv

# Activate on Windows PowerShell
.\.venv\Scripts\Activate.ps1
```

## 6. Jupyter Notebook & Core Data Stack Installation

With your dedicated virtual environment active, install Jupyter Notebook, JupyterLab, and the standard KPMG analytical library stack:

### Command Execution:

```
pip install --upgrade pip
pip install jupyter notebook jupyterlab ipykernel
pip install numpy pandas scikit-learn matplotlib seaborn openpyxl requests sqlalchemy
```

### Registering the Environment Kernel in Jupyter:

To allow switching between multiple project environments within the same Jupyter session:

```
python -m ipykernel install --user --name=kpmg_analytics --display-name="Python (KPMG Analytics)"
```

## 7. Verification & Launching Jupyter

### 1. Launch Jupyter Notebook:

In your terminal / Anaconda Prompt (within your engagement working directory), run:

```
jupyter notebook
```

2. Browser Interface: The local web server will automatically initialize and open <http://localhost:8888> in your default browser (Microsoft Edge / Google Chrome).

### 3. Sanity Test Script:

Create a new notebook with the Python (KPMG Analytics) kernel and execute the verification cell:

```
import sys
import pandas as pd
import numpy as np

print(f"Python Version: {sys.version}")
print(f"Pandas Version: {pd.__version__}")
print(f"NumPy Version: {np.__numpy_version__ if hasattr(np, '__numpy_version__') else np.__version__}")
print("[OK] KPMG Analytics Environment successfully configured!")
```

## 8. Governance, Data Privacy & Information Security Protocols

Adherence to KPMG Global Information Security Policies (GISP) is mandatory:

- Client Data Protection: Never store unencrypted client data, Personally Identifiable Information (PII), or confidential artifacts in local Git repositories or unapproved cloud drives.
- Repository Management: When using Azure DevOps / GitLab / GitHub Enterprise, always maintain a robust .gitignore excluding .ipynb\_checkpoints/, \*.csv, \*.xlsx, credentials, and local virtual environment folders (.venv/).

- Code Reviews: Ensure notebooks converted to production scripts undergo peer review and static analysis.

## 9. Troubleshooting & Support Contacts

| Issue / Symptom                     | Potential Root Cause                         | Recommended Action                                                  |
|-------------------------------------|----------------------------------------------|---------------------------------------------------------------------|
| SSL: CERTIFICATE_VERIFY_FAILED      | Zscaler / Proxy SSL interception             | Update root certificates or append --trusted-host flags in pip.ini. |
| Command 'jupyter' not found         | Virtual environment inactive or PATH missing | Run conda activate kpmg_analytics prior to running Jupyter.         |
| Kernel not appearing in Jupyter     | Kernel spec not registered                   | Execute python -m ipykernel install --user --name=<env_name>.       |
| Package conflict during pip install | Incompatible dependency versions             | Use clean virtual environments for each client engagement.          |

For elevated technical support, submit a ticket via the KPMG IT Service Portal (ServiceNow) under Developer Tooling & Analytics Support.

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